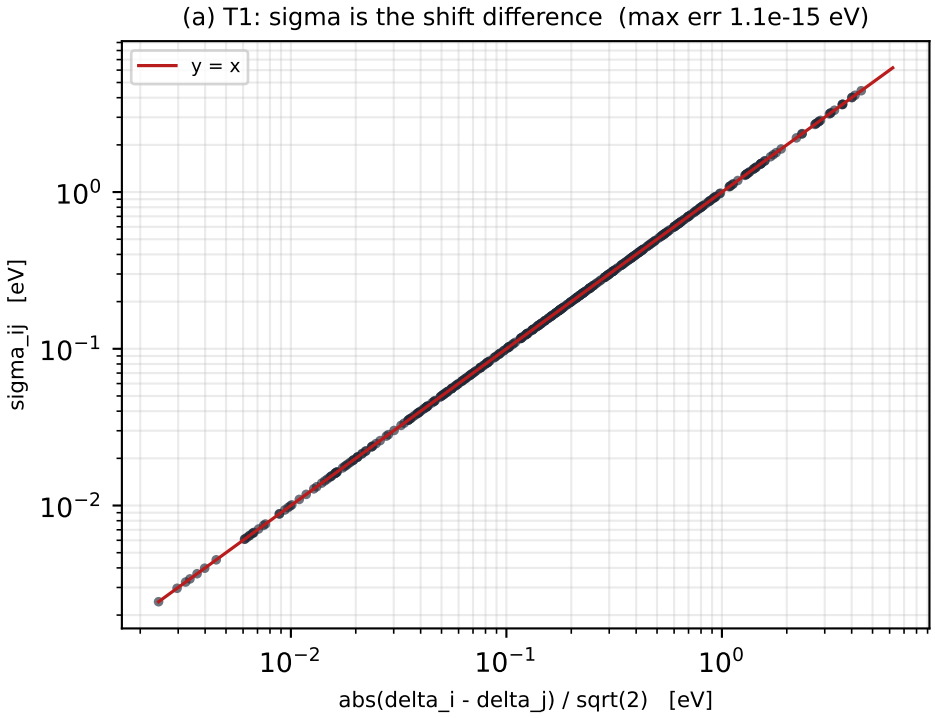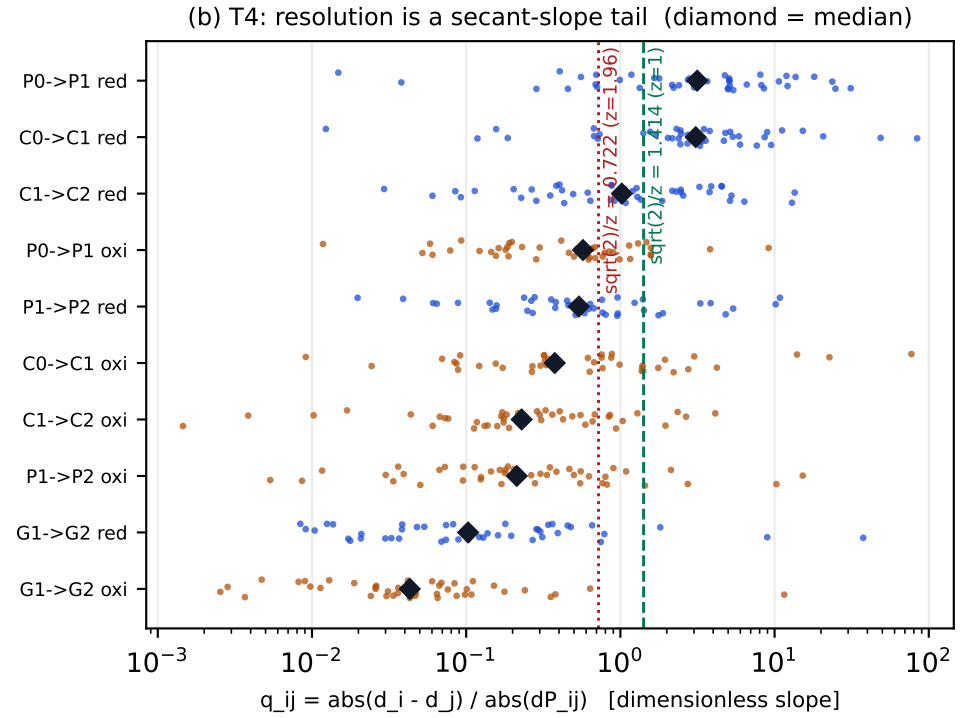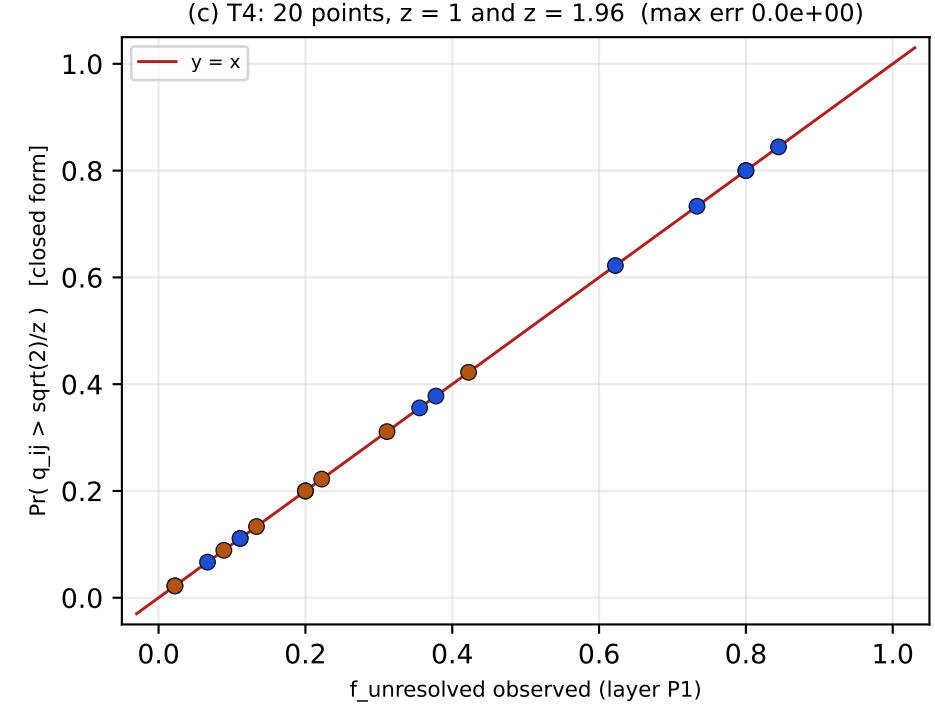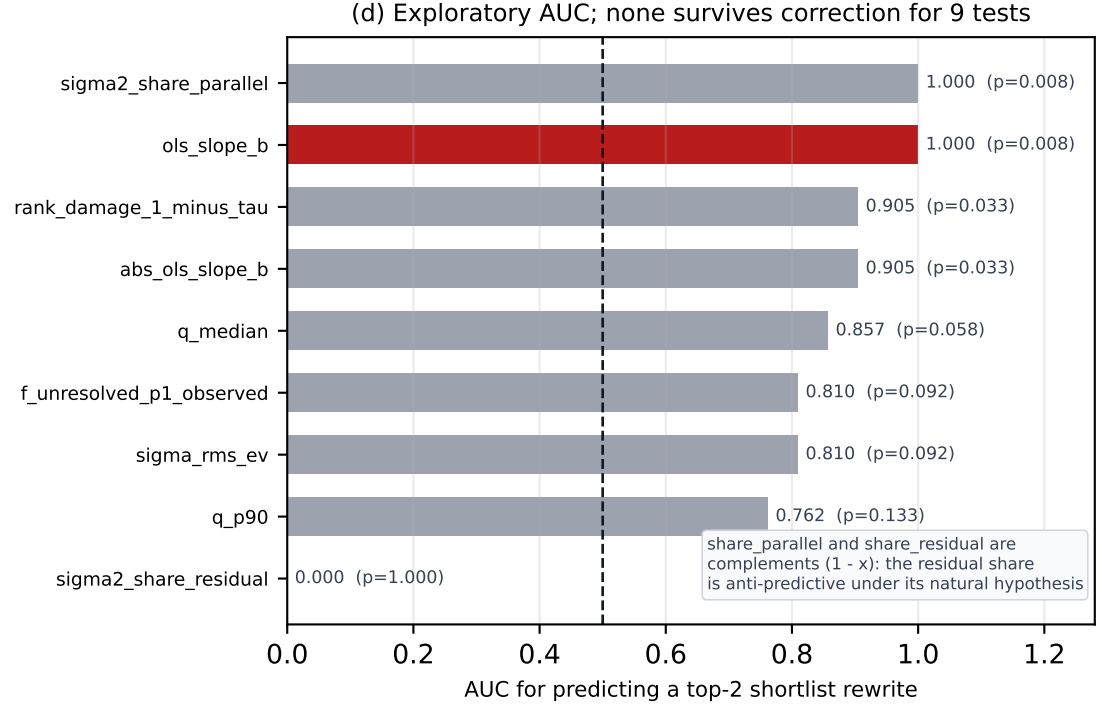Two-arm disagreement convention: algebraic identities and exploratory prediction